Department of Electrical Engineering		
National Institute of Technology Kurukshetra		
Course : AC Machine	Semester: 3rd	Total Questions : 4
Course Code : EEPC201	Date : 24/09/2025	Question to attempt : 3
Program : B.tech (EE)	Time: 1:00-1:50 PM	Max. Marks : 15
Exam : Mid-Sem 1	Centre : E101,E102	Max. Time : 50 min

Note : Assume the necessary data if required

Q. No	Question	Marks
1)	(a) Draw the diagram of a single-phase Shell type transformer to show all windings. (b) Draw the phasor diagram of the transformer under 1. Resistive load 2. lagging p.f. load.	5
2)	(a) Explain the differences between the approximate equivalent circuit and exact circuit of a single phase transformer. (b) Define all the losses of the transformer in no load and full load condition.	5
3)	A short circuit test, when performed on the H.V. side of 10 kVA , 2000/400v single phase transformer gave the following data : 60V, 4A, 100watts If the L.V side is delivering full load (or rated) current at 0.8 p.f. lag and at 400v, find the voltage applied to H.V. side.	5
4)	(a) Why does the flux in the core of the transformer remain constant? (b) A 3300/330v, 50Hz, single-phase transformer has an exciting current of 0.6A and a core loss of 360 watts when its H.V. side is energized at rated voltage. Calculate two components of exciting current.	5